ECO2010H1F

Problem Set 11 Solutions

Question 1

(a) Define the Euler residual as a function of θ,

$$m_{t+1}(\theta) \equiv 1 - \beta\, R_{t+1} \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma}.$$

The unconditional population moments are

$$\mathbb{E}\big[Z_t\, m_{t+1}(\theta)\big] = \mathbf{0}_3.$$

Writing this component-wise with $Z_t = [1,\ \Delta \log C_t,\ W_t]'$,

$$h(\theta) \;=\; \begin{bmatrix} \mathbb{E}\big[m_{t+1}(\theta)\big] \\ \mathbb{E}\big[\Delta \log C_t \cdot m_{t+1}(\theta)\big] \\ \mathbb{E}\big[W_t \cdot m_{t+1}(\theta)\big] \end{bmatrix} \;=\; \mathbf{0}$$

(b) Using the $T-1$ observable Euler pairs,

$$\begin{aligned} \widehat{h}_T(\theta) \;&=\; \frac{1}{T-1} \sum_{t=1}^{T-1} Z_t\, m_{t+1}(\theta) \\ &=\; \frac{1}{T-1} \sum_{t=1}^{T-1} \begin{bmatrix} m_{t+1}(\theta) \\ \Delta \log C_t \cdot m_{t+1}(\theta) \\ W_t \cdot m_{t+1}(\theta) \end{bmatrix}. \end{aligned}$$

(c) We have $k. = 2$ and $q = 3$, so the model is overidentified.

(d) Define

$$J_T(\theta) \;=\; \widehat{h}_T(\theta)'\, W_T\, \widehat{h}_T(\theta),$$

with W_T positive definite. Since the population matrix

Efficient GMM:

1. Minimize $J_T(\theta)$ with a convenient $W_T^{(0)}$ (e.g. the identity matrix) to obtain a preliminary estimate $\tilde{\theta}$.

2. Compute residuals $\tilde{m}_{t+1} = m_{t+1}(\tilde{\theta})$ and form the moment series $Z_t \tilde{m}_{t+1}$.

3. Estimate the asymptotic covariance $\hat{S}_T$ of the moment conditions
4. Set $W_T = \hat{S}_T^{-1}$.
5. Re-minimize $J_T(\theta)$ with W_T to obtain $\hat{\theta}_{\mathrm{GMM}}$.

(e) W_t is an extra predetermined instrument that does not enter the structural Euler equation directly. Adding W_t can reduce the asymptotic variance of the GMM estimator (with the optimal weight) relative to using fewer instruments.

Question 2

We have

$$\begin{aligned} L(\lambda|y) &= \frac{\lambda^y \exp(-\lambda)}{y!} \\ \ell(\lambda|y) &= y \ln \lambda - \lambda - \ln(y!) \\ \frac{\partial \ell(\lambda|y)}{\partial \lambda} &= y\lambda^{-1} - 1 \\ \frac{\partial^2 \ell(\lambda|y)}{\partial \lambda^2} &= -y\lambda^{-2} \end{aligned}$$

and therefore the Fisher information is obtained as

$$-E\left[-y\lambda^{-2}\right] = \lambda^{-2}E\left[y\right] = \lambda^{-2}\lambda = \lambda^{-1}$$

For a sample of n observations, the likelihood and log-likelihood functions are

$$\begin{aligned} L(\lambda|\mathbf{y}) &= \prod_{i=1}^{n} \frac{\lambda^{y_i} \exp(-\lambda)}{y_i!} \\ \ell(\lambda|\mathbf{y}) &= \ln \lambda \sum_{i=1}^{n} y_i - n\lambda - \sum_{i=1}^{n} \ln(y_i!) \end{aligned}$$

yielding the F.O.C.

$$\frac{\partial \ell(\lambda|\mathbf{y})}{\partial \lambda} = \lambda^{-1} \sum_{i=1}^{n} y_i - n = 0$$

resulting in

$$\widehat{\lambda}_{MLE} = n^{-1} \sum_{i=1}^{n} y_i$$